BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Full-Course Synoptic Assessment

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the IDE editor used for?

2. In OCR ERL, what is 2 ^ 3?

A) 6 B) 8
C) 9 D) 23

3. "Casting guarantees that every possible input string becomes a valid integer." Correct this. Explain why.

4. A warning needs high temperature OR high pressure. "OR is false when both inputs are true." Correct this. What happens when both are high?

Today's objectives

Complete an independent 45-minute synoptic assessment. Answer questions from both OCR J277 papers.

Key vocabulary - Read, cover, recall

Compiler
Translates a whole program before running.

Interpreter
Executes source statements during translation.

Trace table
Shows changes to variable values.

Truth table
Lists all Boolean input combinations.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Assessment guidance	Answer each question without notes. Read the number of marks and use appropriate command words.
Paper 1	Recall architecture, storage, networking, security, systems software and legal impacts.
Paper 2	Use computational thinking, algorithms, programming, Boolean logic and language knowledge.
Review	Check calculations, code traces and linked explanations before submission.

I Do - One worked example

Practice only: Explain why storing active programs in RAM helps the computer operate. [2 marks]

1. RAM is used for active programs and working data.

2. RAM holds currently running instructions/data [1].

3. The CPU can access this working information quickly when carrying out instructions [1].

4. Two connected points explain a purpose; this practice example is not part of the timed paper.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Practice only: A school wants to keep private records secure. Explain one benefit of encryption. [2 marks]

2. Practice only: Break down a library fine-calculation program into three smaller tasks. [3 marks]

3. Practice only: Trace total = 5; total = total + 2; print(total). State values and output. [3 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Assessment coding Q1: predict and edit this OCR Exam Reference Language program.

```
unitPrice = 7
quantity = 3
total = unitPrice * quantity
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Answer the full assessment paper in 45 minutes.

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A compiler translates all the source code before execution. ☐ T ☐ F

2. Answer each question without notes ☐ T ☐ F

3. Recall architecture, storage, networking, security, systems software and legal impacts. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Writing and editing source code, often with helpful formatting or highlighting. ☐

Q2 B - 8 ☐

Q3 Convert the text to a suitable numeric type before doing numeric calculations. ☐

Q4 OR is true if at least one input is true, including both. AND needs both true. NOT reverses a value. ☐

Code First - Model

First output: 21 [1]. At quantity = 4: 28 [1].

Possible correct extension:
fee = 2
print(total + fee)

Award [1] for fee/updated addition and [1] for printing the correct updated total of 30. This model assumes the revised quantity is 4.

We Do - Check the model

1. Encryption turns readable data into ciphertext [1], making it difficult to read without the correct key [1]. ☐

2. Input the number of overdue days [1]; calculate fine using a daily rate [1]; display the total [1]. ☐

3. After first assignment total is 5 [1]; after addition total is 7 [1]; output is 7 [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Mark using the separate assessment answer key after the paper. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: